

Bitcoins and Cryptocurrencies

Note: most of the slides used in this course are derived from those available for the book “Bitcoins and Cryptocurrencies Technologies – A Comprehensive Introduction”, Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller & Steven Goldfeder, 2016, Princeton University Press.

Lecture 6

Bitcoin and anonymity

Anonymity basics

Some say Bitcoin provides anonymity

“ Bitcoin is a secure and anonymous digital currency ”

— WikiLeaks donations page

Others say it doesn't

“Bitcoin won't hide you from the NSA's prying eyes”

— Wired UK

What do we mean by anonymity?

- Literally: ***anonymous*** = '*without a name*'
 - Two interpretations:
 - 1 – without using your real name
 - 2 – without using any name at all
- Bitcoins: Addresses are public key hashes rather than real identities
 - By first interpretation, Bitcoin is anonymous
 - By second interpretation, it is not anonymous! Bitcoin addresses → pseudo-identity
 - Computer scientists call this ***pseudonymity***

Pseudonymity vs anonymity in forums

Reddit: pick a long-term pseudonym

vs.

4Chan: make posts with no attribution at all

Anonymity in computer science

Anonymity = pseudonymity + **unlinkability**



Different interactions of the same user with the system should not be linkable to each other (and to the user's real identity)

Question: Why is unlinkability needed?

- Answer: Pseudonymity is not enough for “privacy” in Bitcoins!
- Bitcoin Block chain (and transactions) is public
- Linking Bitcoin addresses to real identities is easy
 1. Many Bitcoin services require real identity
 2. Linked profiles can be deanonymized by a variety of side channels

Defining unlinkability in Bitcoin

1. Hard to link together different addresses of the same user
2. Hard to link together different transactions made by the same user
3. Hard to link sender of a payment to its recipient

Quantifying Bitcoin anonymity

Complete unlinkability (among all addresses/transactions) is hard → lets' try to get something more feasible

Anonymity set: the crowd that one attempts to blend into

Anonymity set (for a Bitcoin transaction): set of transactions that an **adversary** cannot distinguish from the transaction in question

To calculate anonymity set:

- define adversary model
- reason carefully about: what the adversary knows, does not know, and cannot know

Alternate measures of Bitcoin anonymity

Taint Analysis: Give high “taint” score (to S - R pair) if Bitcoins sent by an address S always end up at an address R

Not a good measure of Bitcoin anonymity:

- Makes implicit assumptions about the adversary’s calculations regarding linking pairs of addresses
- Adversary may use alternate techniques for linking address pairs – e.g., timing analysis or wallet software idiosyncrasies.

Why anonymous cryptocurrencies?

Block chain based currencies are totally, publicly, and permanently traceable

Without anonymity, privacy is much worse than traditional banking!

Two motivations:

1. Achieve privacy level at least equivalent to traditional banking
2. If possible, go beyond the privacy level offered by traditional banking based solutions

Ethical concerns about anonymity

Legitimate “goods”: may prevent learning
someone’s salary information or an organization’s
private business dealing

Legitimate “worries” : money laundering
Bottleneck: moving large flows into and out of
Bitcoin (“cashing out”)

Can we keep only the good uses?

Common conundrum in computer security and privacy:

uses that are very different morally are pretty much the same technologically

Similar dilemma: Tor

Anonymous communication
network

Sender and receiver of
message unlinkable

Used by:

- Normal people
- Journalists & activists
- Law enforcement
- Malware
- Child pornographers

Funded by (among others):
U.S. State Department

Anonymous e-cash: history

David Chaum, 1982

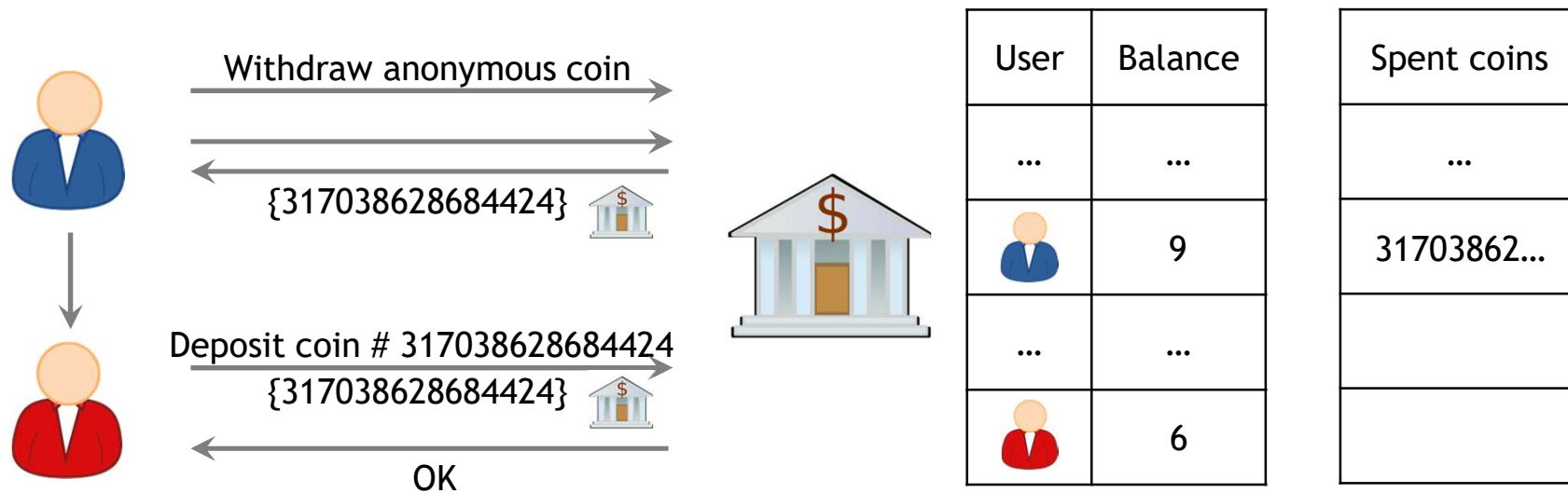
Blind signature:

two-party protocol to create digital signature
without signer knowing the input



Crypto
magic

Anonymous e-cash via blind signatures



Bank cannot link the two users

Anonymity & decentralization: in conflict

- Interactive protocols with bank are hard to decentralize
- Decentralization often achieved via public traceability to enforce security

How to de-anonymize Bitcoin?

Bitcoin

Bitcoin is a secure and anonymous digital currency. Bitcoins cannot be easily tracked back to you, and are safer and faster alternative to other donation methods. You can send BTC to the following address:

13DFamCvSxG8EG16VyXzdpfqxyooifswYx 

Various sites offer a service to exchange other currency to/from Bitcoins. There are also services allowing trades of goods for Bitcoins. Bitcoins are not subject to central regulations and are still gaining value. To learn more about Bitcoins, visit the website (<http://bitcoin.org>) or read more on [Wikipedia](#).

To generate a new, private address for your donation, click the refresh button above.



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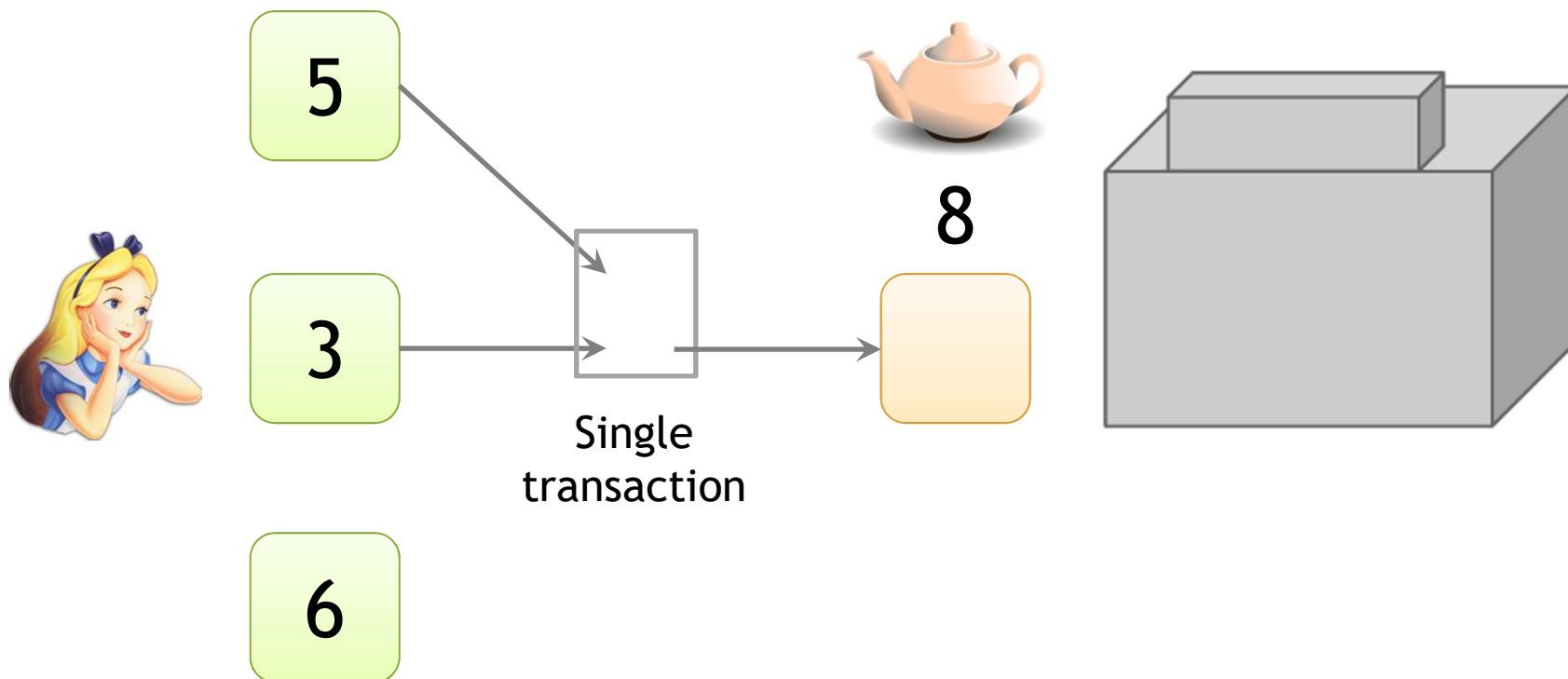


Trivial to create new address

Best practice: always receive at fresh address

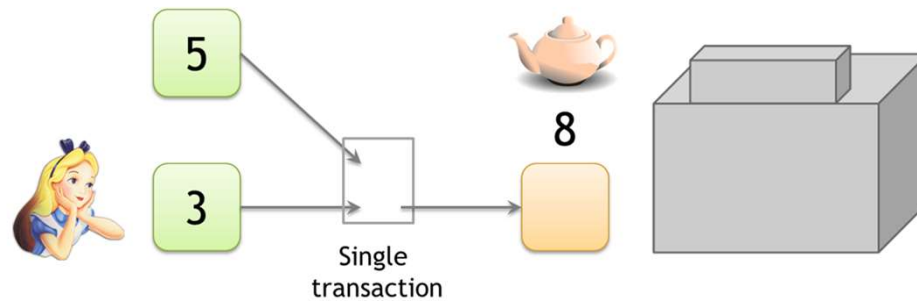
So, unlinkable?

Alice buys a teapot at Big box store



Linking addresses

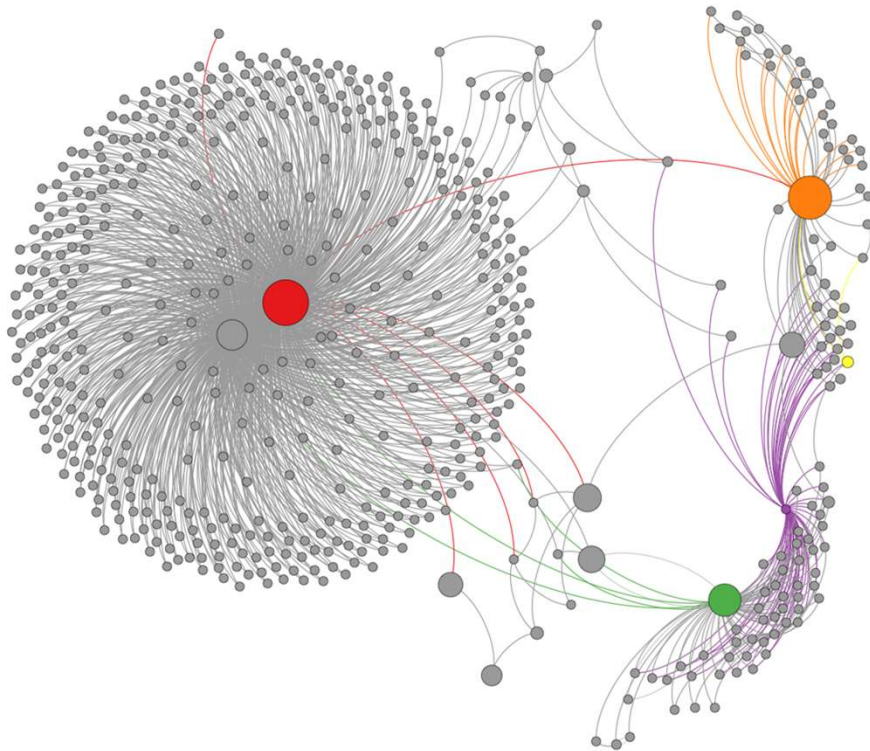
Shared spending is evidence of joint control



Addresses can be linked transitively

Clustering of addresses in Bitcoin (or other cryptocurrencies) is the process of grouping multiple addresses that are likely controlled by the same user or entity.

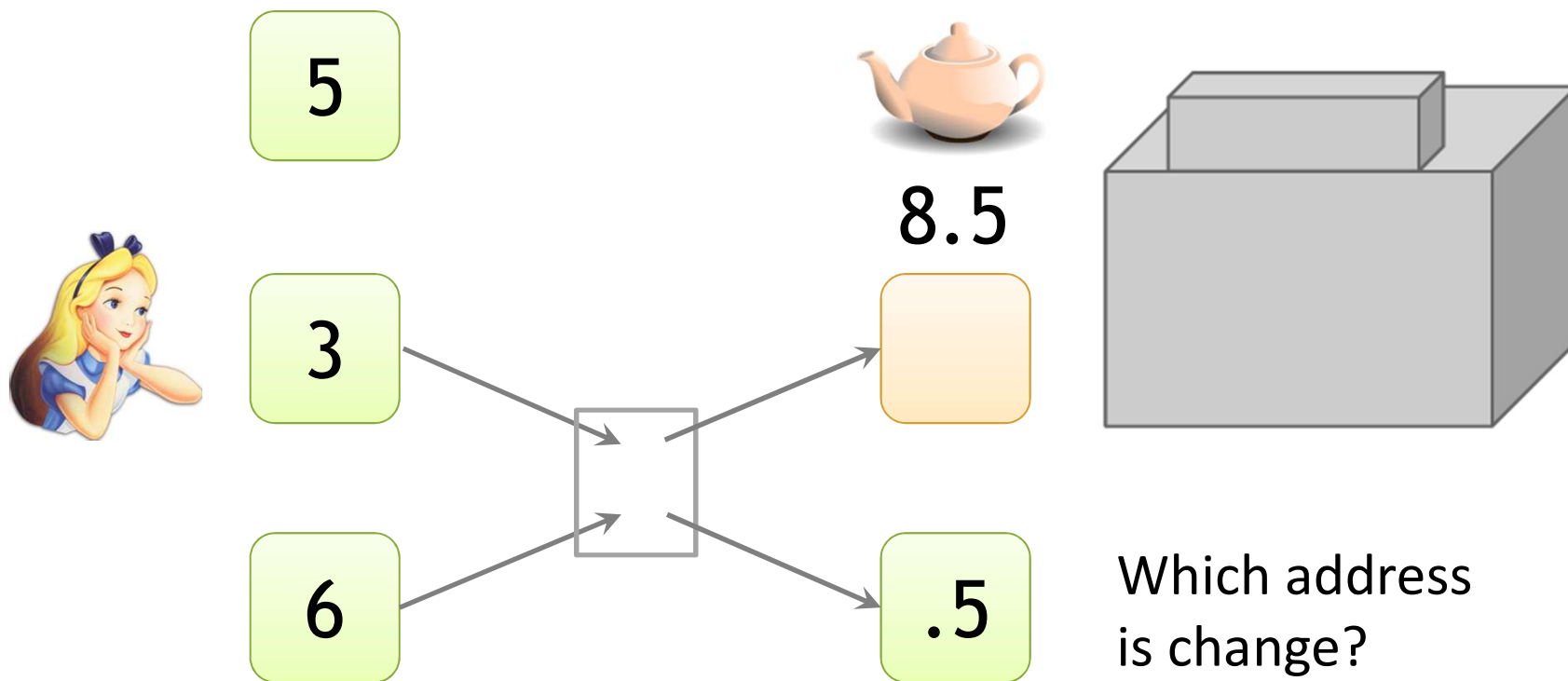
Clustering of addresses



*An Analysis of Anonymity in
the Bitcoin System*

F. Reid and M. Harrigan
PASSAT 2011

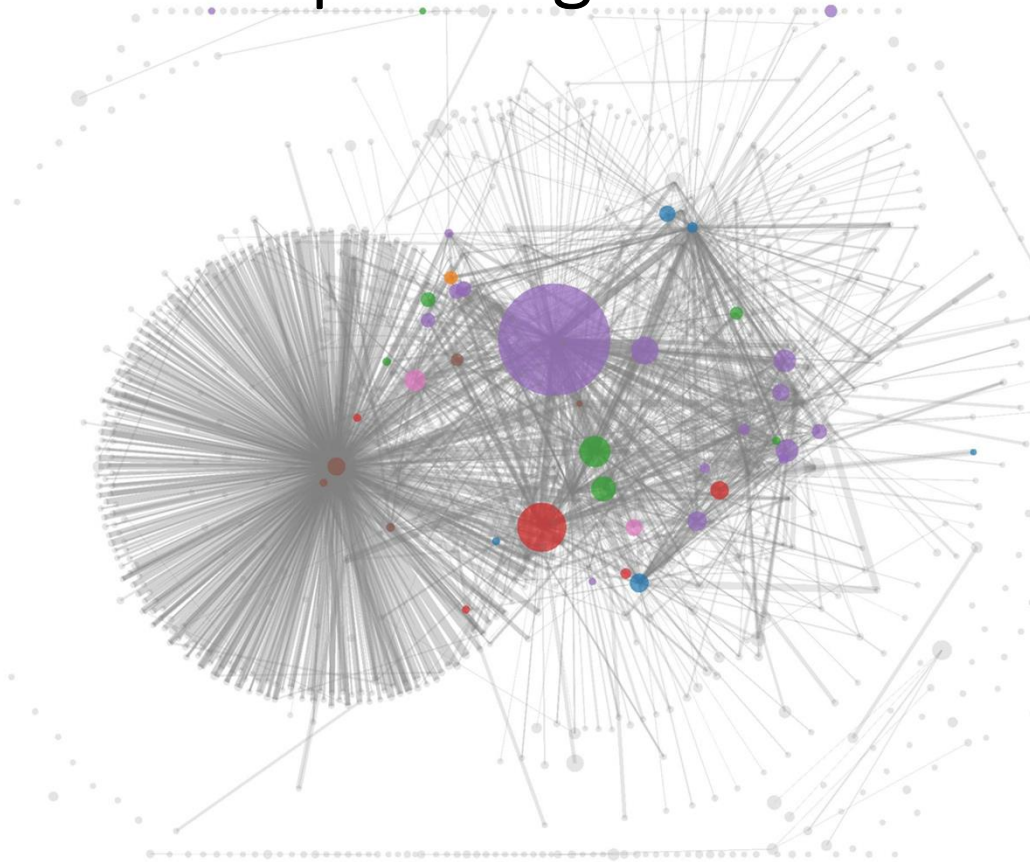
Change addresses



“Idioms of use”

- Implementation details or Idiosyncratic features of wallet software that can be used to design heuristics for identifying (linking) **change addresses**
- For example, most wallets generate a fresh address (never appeared on the Blockchain) when a change address is required
- Bitcoin-Qt library bug → change output always first output in a transaction

Shared spending + idioms of use



*A Fistful of Bitcoins:
Characterizing Payments
Among Men with No Names*

S. Meiklejohn et al.
IMC 2013

To tag service providers: transact!



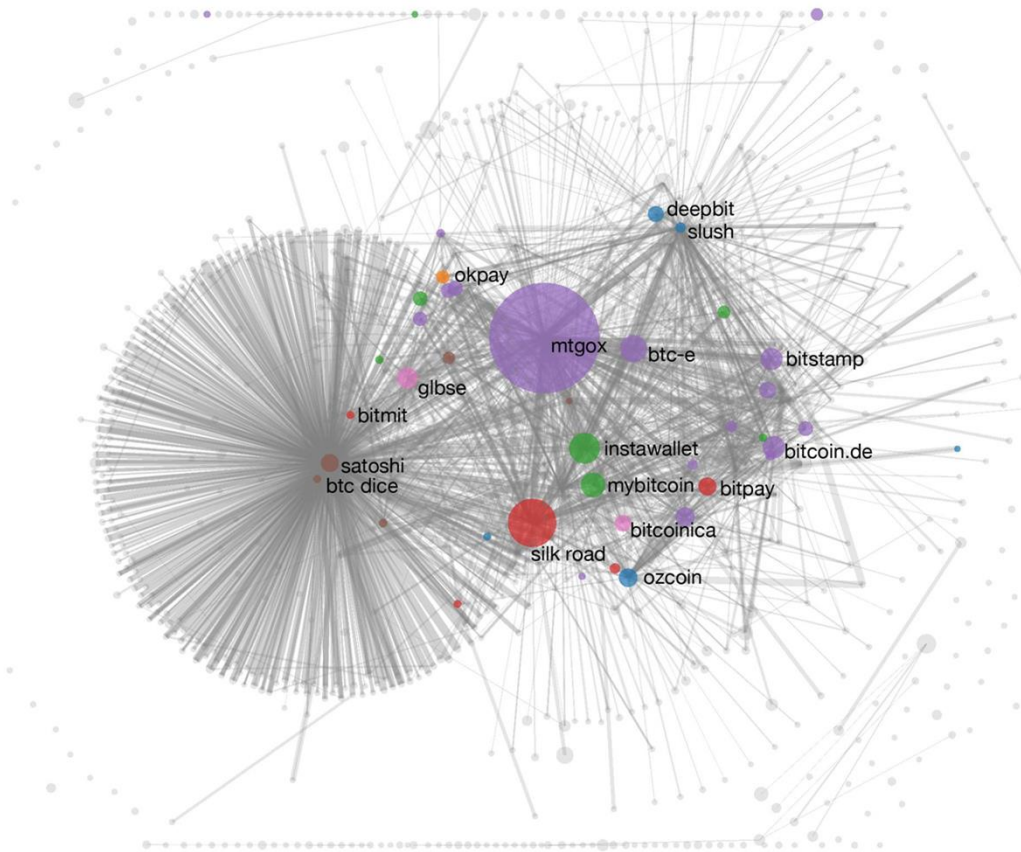
A Fistful of Bitcoins: Characterizing Payments Among Men with No Names

S. Meiklejohn et al.

344 transactions

- Mining pools
- Wallet services
- Exchanges
- Vendors
- Gambling sites

Shared spending + idioms of use



*A Fistful of Bitcoins:
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S. Meiklejohn et al.

From services to users

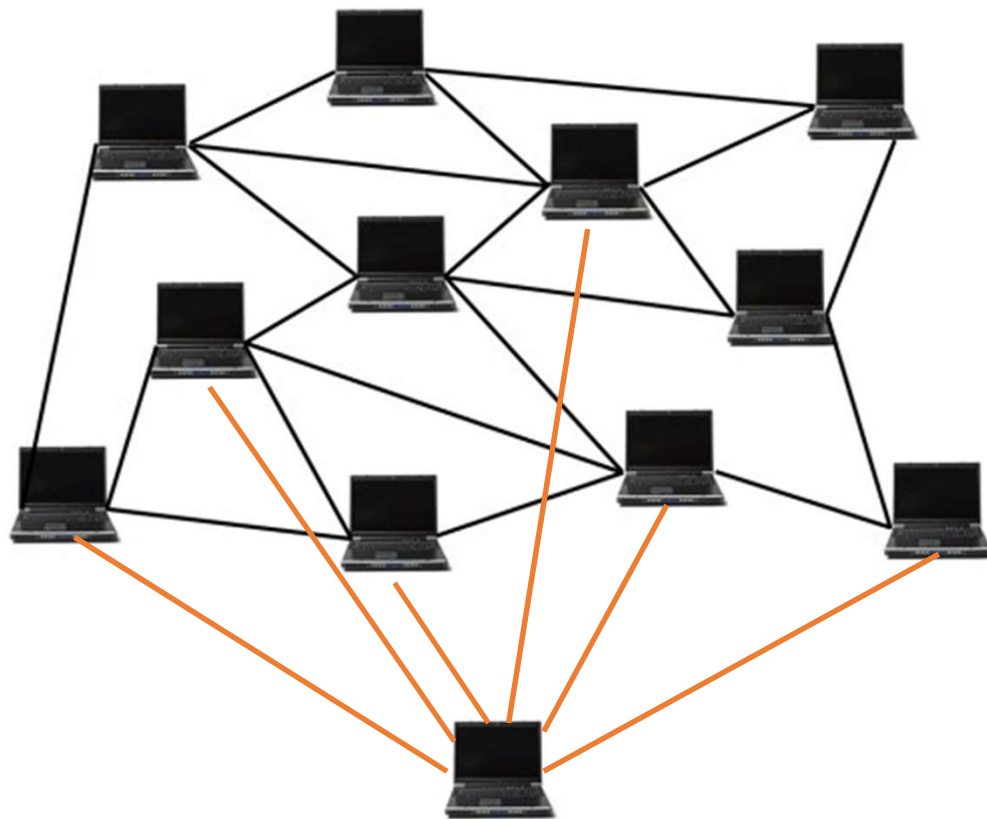
1. High centralization in service providers

Most flows pass through one of these — in a traceable way

2. Address — identity links in forums

for identifying
individuals

Network-layer de-anonymization



“The first node to inform you of a transaction is probably the source of it”

Dan Kaminsky
Black Hat 2011 talk

Solution: use Tor

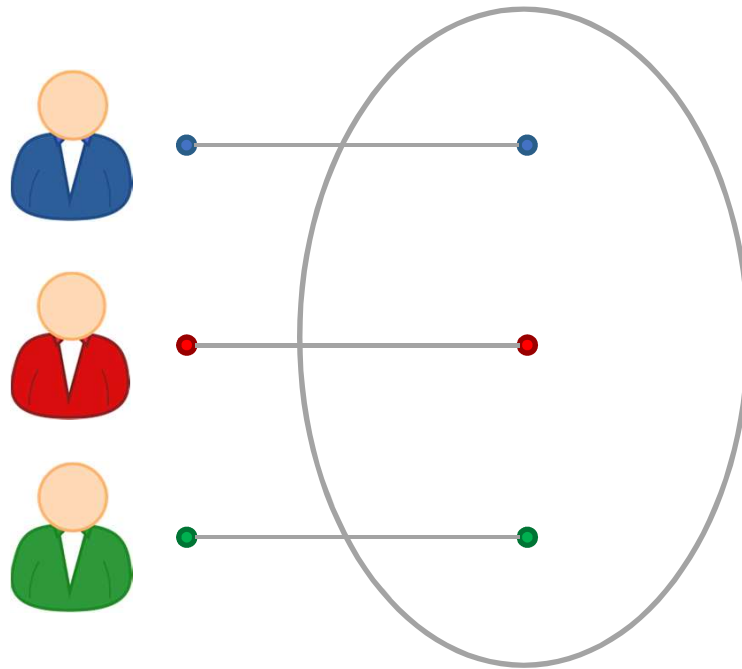
Caveat: Tor is intended for low-latency activities such as web browsing

Mix nets might provide better anonymity

BUT Tor is what's deployed and works

Mixing

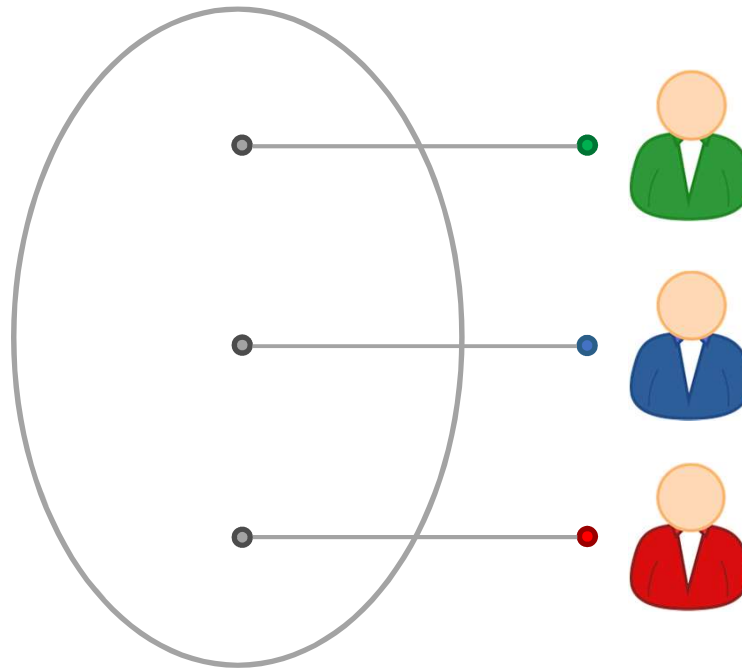
To protect anonymity, use an intermediary



To protect anonymity, use an intermediary

Online wallets do
this

Do they provide
anonymity?!





Study Suggests Link Between Dread Pirate Roberts and Satoshi Nakamoto

By **JOHN MARKOFF** NOVEMBER 23, 2013 6:13 PM [23 Comments](#)



E-MAIL



FACEBOOK



TWITTER



SAVE

Two Israeli computer scientists say they may have uncovered a puzzling financial link between Ross William Ulbricht, the [recently arrested](#) operator of the Internet black market known as the Silk Road, and the secretive inventor of bitcoin, the anonymous online currency, used to make Silk Road purchases.



Researchers Retract Report That Linked Bitcoin Creator and Silk Road

By **JOHN MARKOFF** NOVEMBER 27, 2013 12:45 PM [6 Comments](#)



E-MAIL



FACEBOOK



TWITTER



SAVE



MORE

Two Israeli computer scientists who over the weekend published a paper describing a financial connection between the Bitcoin peer-to-peer transaction system and the operator of Silk Road, an Internet black market, have backed away from the claim after an independent security researcher took responsibility for the puzzling account that generated the transfer.

Online wallets for anonymity

Reputable, often regulated, businesses

- Typically require identity, keep records → no anonymity w.r.t. wallet service
- Users trust them with their bitcoins → keep them for longer → bigger anonymity set w.r.t. everyone else

Dedicated mixing services

- Promise not to keep records
- Don't ask for your identity
- Essentially swaps a user's coin (address) with other users' coins (addresses)
- Relatively small anonymity set (only comprise of those users who use the mixing service at that instant)!

Rest of this lecture:

assume a user for whom the trust requirements and anonymity properties of online wallets are unacceptable

Mixing: terminology

Mix vs. mixer

Another term: laundry

Won't use in this lecture

Principles for mixing services

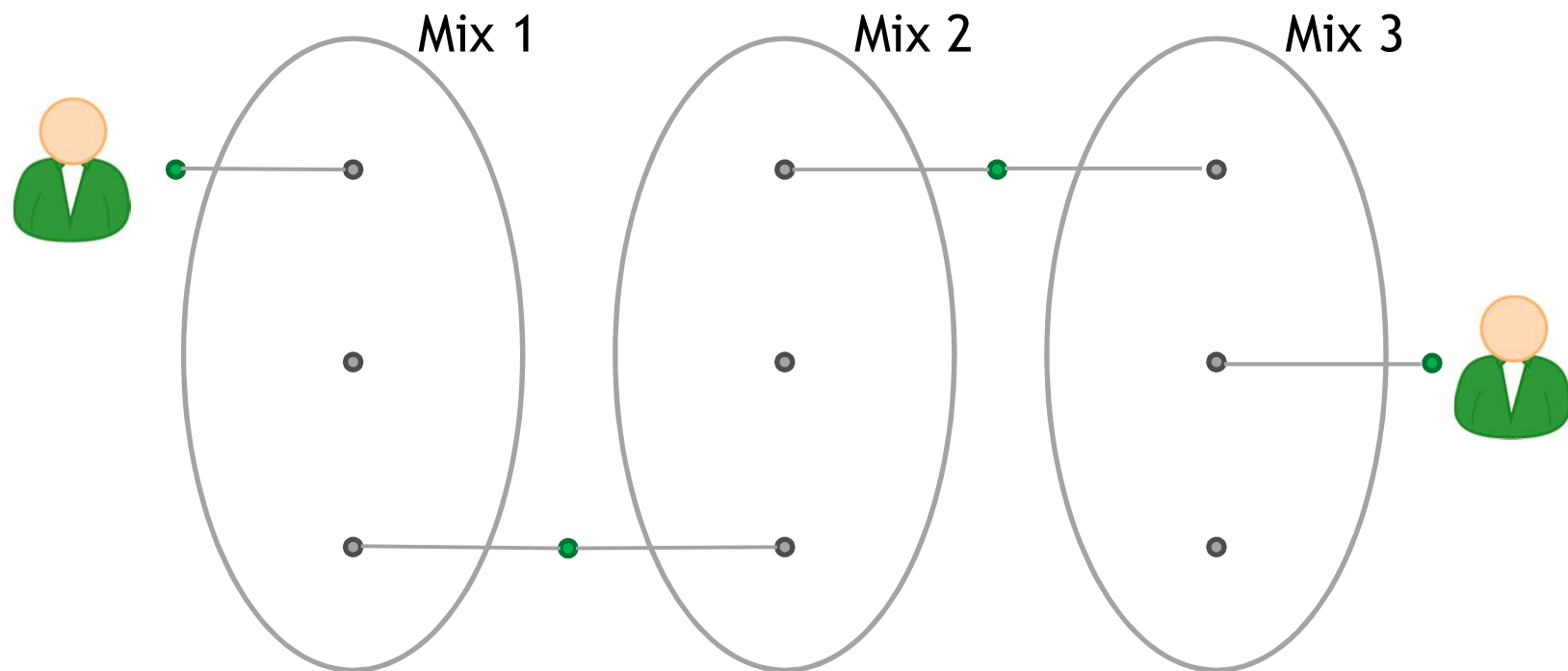
1. Use a series of mixes

Mixes should implement a standard API to make this easy

Mixcoin: Anonymity for Bitcoin with accountable mixes

J. Bonneau et al.
Financial Cryptography
2014

Series of mixes



Principles for mixing services

2. Uniform transactions

In particular: all mix transactions must have the same value!

“Chunk size”

Mixcoin: Anonymity for Bitcoin with accountable mixes

J. Bonneau et al.
Financial Cryptography
2014

Principles for mixing services

3. Client side must be automated

Desktop wallet software

Mixcoin: Anonymity for Bitcoin with accountable mixes

J. Bonneau et al.
Financial Cryptography
2014

Principles for mixing services

4. Fees must be all-or-nothing

Probabilistic fees:

0.1% mixing fee =
mix will swallow chunk
with 0.1% chance

*Mixcoin: Anonymity for
Bitcoin with accountable
mixes*

J. Bonneau et al.
Financial Cryptography
2014

Current mixes follow none of these principles

Remaining problem: trusting mixes

1. Stay in business, build up reputation
2. Users can test for themselves
3. Cryptographic “warranties”

Currently no reputable dedicated mix

Caution: Mixing services may themselves be operating with anonymity. As such, if the mixing output fails to be delivered or access to funds is denied there is no recourse. Use at your own discretion.

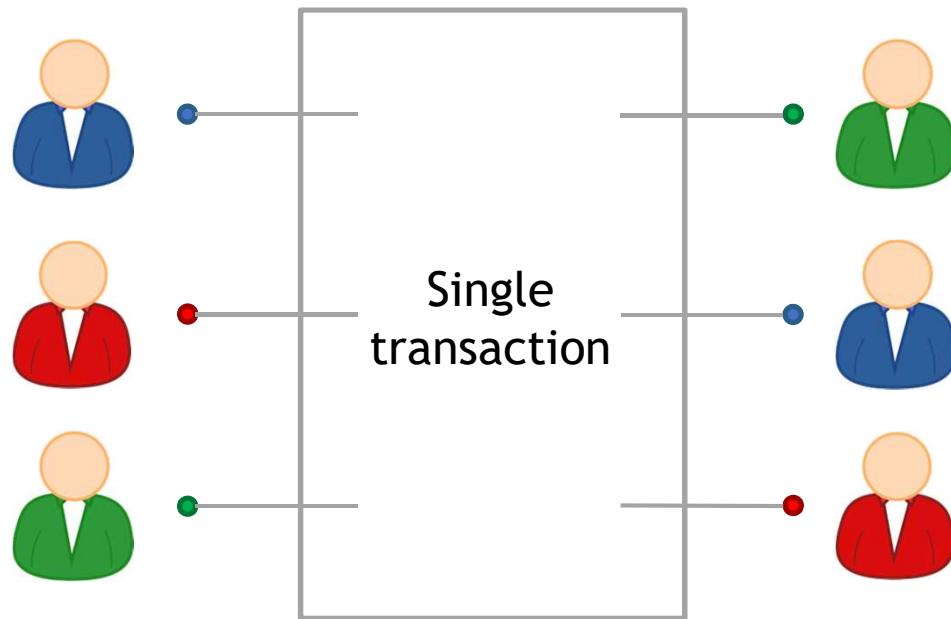
— Bitcoin Wiki

Decentralized mixing

Why decentralized mixing?

- No bootstrapping problem
- Theft impossible
- Possibly better anonymity
- More philosophically aligned with Bitcoin

Coinjoin



Each signature is entirely separate

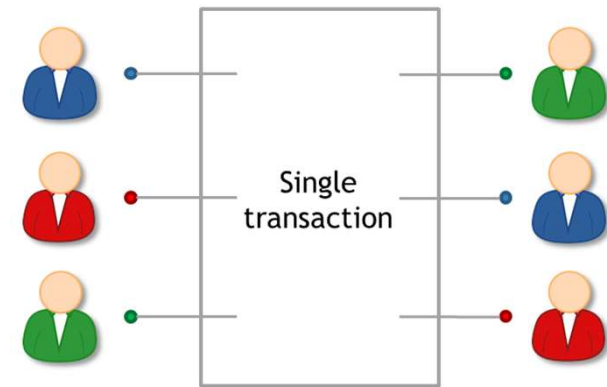
This is 1 mixing round

Mixing principles from before apply on top of basic protocol

Proposed by Greg Maxwell, Bitcoin core developer

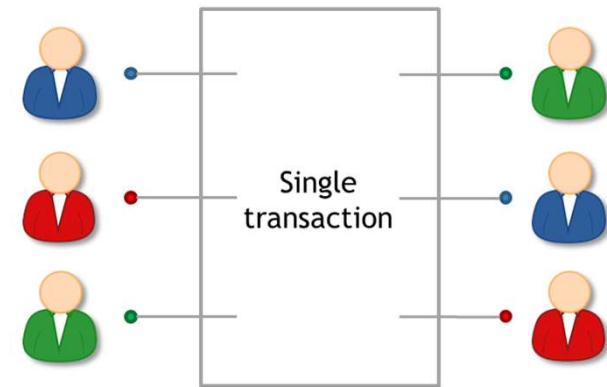
Coinjoin algorithm

1. Find peers who want to mix
2. Exchange input/output addresses
3. Construct transaction
4. Send it around, collect signatures
(Before signing, each peer checks if her output is present)
5. Broadcast the transaction



Coinjoin: remaining problems

- How to find peers
- Peers know your input-output mapping
(This is a worse problem than for centralized mixes)
- Denial of service



Finding peers

- Use an untrusted server to facilitate this process
- Unlike centralized mixes, these servers cannot steal users' funds

Peer anonymity

Strawman solution:

1. exchange inputs
2. disconnect and reconnect over Tor
3. exchange outputs

Better solution:

special-purpose anonymous routing mechanism

Denial of service

Proposed solutions:

- Proof of work
- Proof of burn
- Server kicks out malicious participant
- Cryptographic “blame” protocol

(CoinShuffle: Practical Decentralized Coin Mixing for Bitcoin, T. Ruffing et al., PETS 2014)

High-level flows could be identifying

Example:

Alice receives 43.12312 BTC / week as income

Always immediately transfers 5% to retirement account

Heuristic: merge avoidance

Instead of a single payment transaction

receiver provides multiple output addresses
sender avoids combining different inputs

(Proposed by Mike Hearn)

Zerocoin and Zerocash

Zerocoin: protocol-level mixing

Mixing capability baked into protocol

Zerocoin: Anonymous Distributed E-Cash from Bitcoin

Advantage: cryptographic guarantee of mixing

I. Miers et al.
IEEE S&P 2013

Disadvantage: not currently compatible with Bitcoin

Basecoin and Zerocoin

Basecoin: Bitcoin-like Altcoin

Zerocoin: Extension of Basecoin

- Basecoins can be converted into zerocoins and back
- Breaks link between original and new basecoin

Zerocoins

- A Zerocoin is a cryptographic proof that you owned a Basecoin and made it unspendable
- Miners can verify these proofs
- Gives you the right to redeem a new Basecoin (somewhat like poker chips)

Two challenges

- How to construct these proofs?
- How to make sure each proof can only be “spent” or “redeemed” once with a basecoin?
 - Otherwise, a basecoin could be turned to a zerocoin which could be redeemed into multiple basecoins

Zero-knowledge proofs

A way to prove a statement without revealing any other information



Example:

- “I know an input that hashes to **da39a3ee5e**”
- “I know an input that hashes to some hash in the following set: ... ”

Minting zerocoins

- Zerocoins come in standard denominations
(Let's assume 1 basecoin)
- Anyone can make (or mint) one!
- They have value once put on the block chain
 - That costs 1 basecoin

Minting a zerocoin: “commitment”

- Generate serial number S (eventually made public)
- and random secret r (never public, ensures unlinkability)
- Compute $\text{Commit}(S, r)$

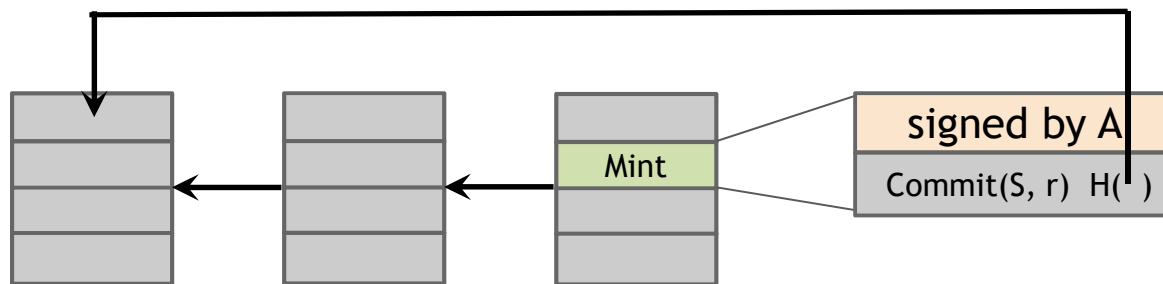


Simplification

Minting a zerocoin

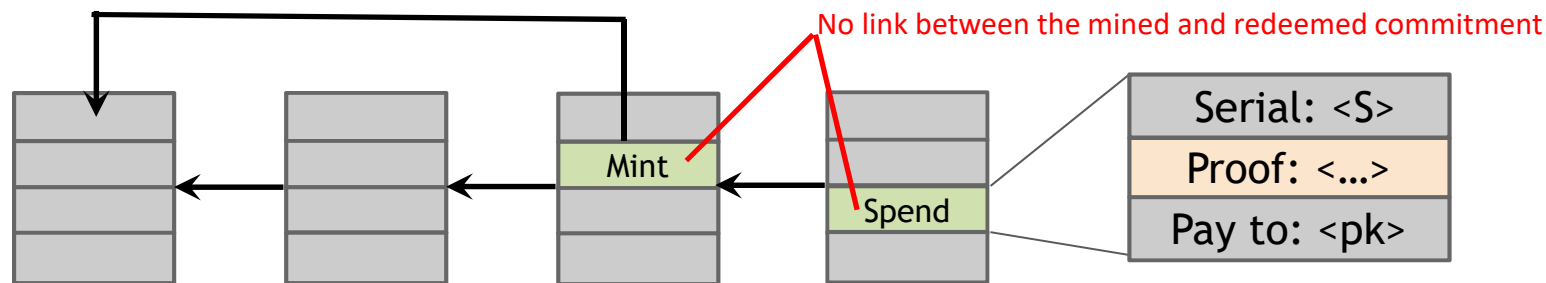
To put $\text{Commit}(S, r)$ on block chain:

Create “Mint” Tx with 1 basecoin as input



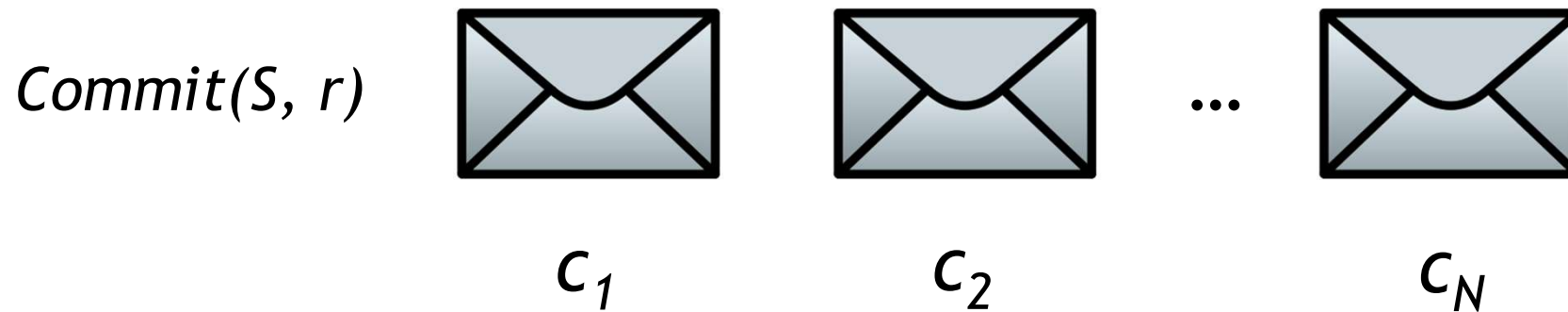
To spend a zerocoin S:

- Create a special “Spend” Tx that contains:
 - Serial number S
 - Zero-knowledge proof that:
“I know a number r such that $Commit(S, r)$ is one of the zerocoins or commitments $\{c_1, c_2, \dots, c_N\}$ in the block chain”
- Miners will verify S hasn’t been spent before. Miners will also verify the above zero-knowledge proof (which establishes the claimant’s ability to open one of the zerocoin commitments without actually opening it)
- Output of the spend Tx now acts as a new basecoin



Zerocoin is anonymous

Since r is secret, no one can figure out *which* zerocoin corresponds to serial number S



Zerocoin is “efficient”

The proof is a giant
disjunction over all
zerocoins

Yet the proof is
relatively small
(logarithmic in N)!

I know r such that

$$\text{Commit}(S, r) = c_1$$

OR

$$\text{Commit}(S, r) = c_2$$

OR

...

OR

$$\text{Commit}(S, r) = c_N$$

Need for a Trusted Setup

- One of the crypto tools (RSA accumulators) requires a one-time trusted setup
- Security relies on difficulty in factorizing a composite N of large primes p and q !
- A trusted party needs to select random p and q , compute $N=pq$, release N and permanently delete p & q
 - *Sociological issue: Can we trust someone to do this?*
- Solution: “Threshold cryptography” – a set of delegates jointly compute N (even if one is honest, difficult to determine p & q)

ZeroCash: Zerocoin without Basecoin

Two differences

- Different crypto for proofs (More efficient)
- Proposal to run system without Basecoin

ZeroCash: Decentralized Anonymous Payments from Bitcoin

E. Ben-Sasson et al.
Usenix Security 2014

ZeroCash: untraceable e-cash

All transactions are zerocoins

Splitting and merging supported

Put transaction value inside the envelope

Ledger merely records existence of transactions

Zerocash: the catch

Random, secret inputs are required to generate public parameters

These secret inputs must then be securely destroyed

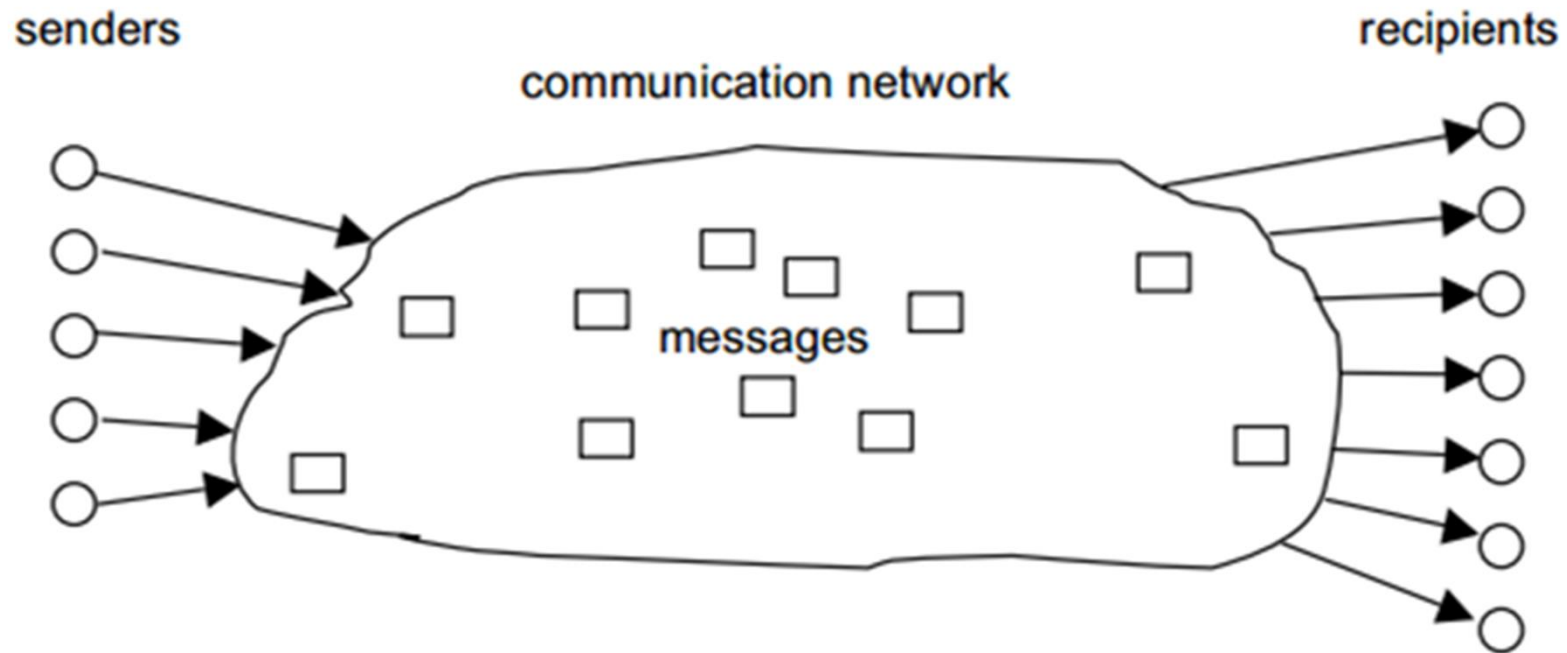
No one can know them (anyone who does can break the system)

5 levels of anonymity

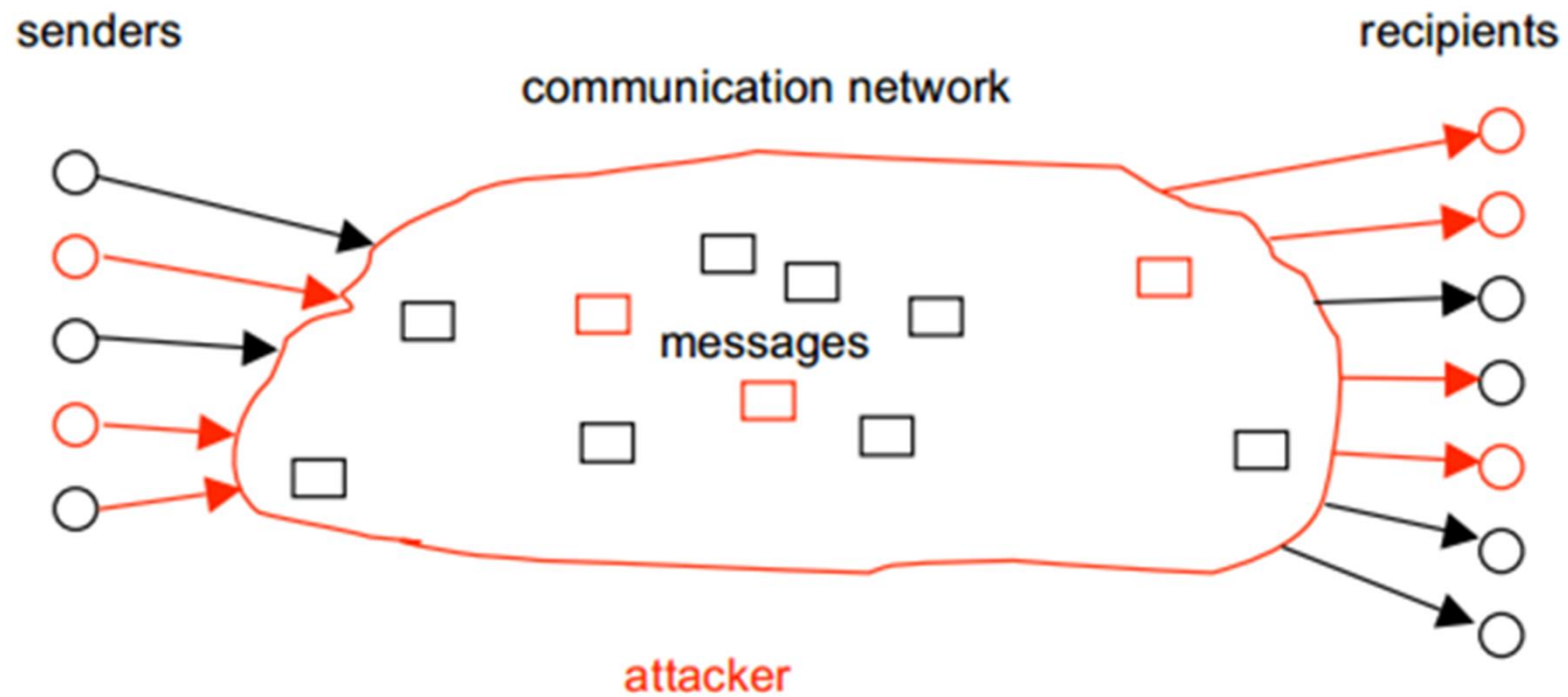
System	Type	Anonymity attacks	Deployability
Bitcoin	Pseudonymous	Tx graph analysis	Default
Single mix	Mix	Tx graph analysis, bad mix	Usable today
Mix chain	Mix	Side channels, bad mixes/peers	Bitcoin-compatible
Zerocoin	Cryptographic mix	Side channels (possibly)	Altcoin
Zerocash	Untraceable	None	Altcoin, tricky setup

Tor and the Silk Road

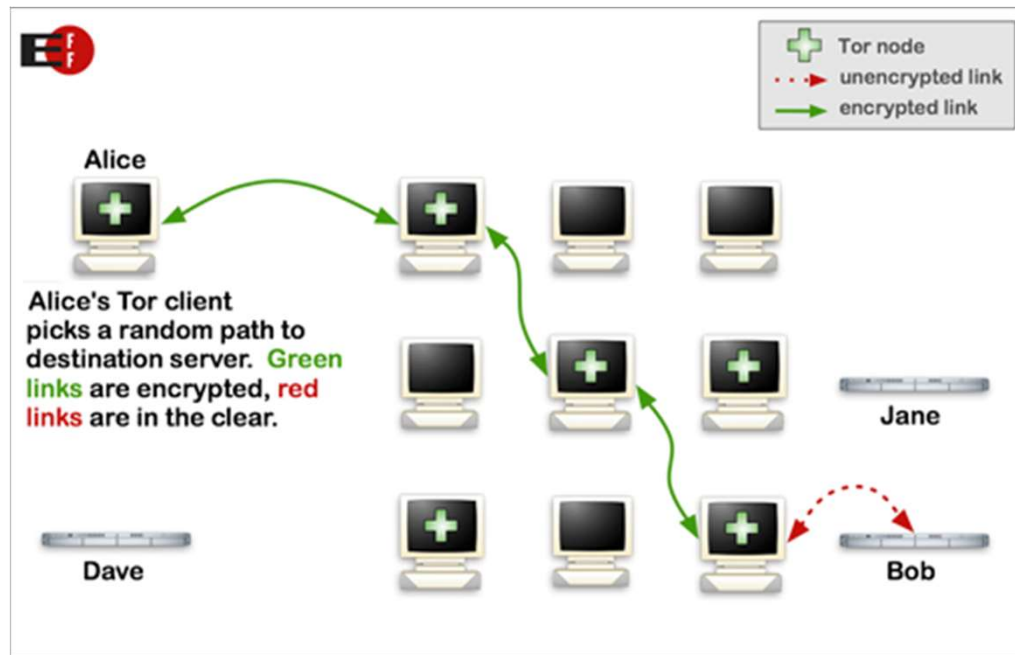
Anonymous communication



Threat model



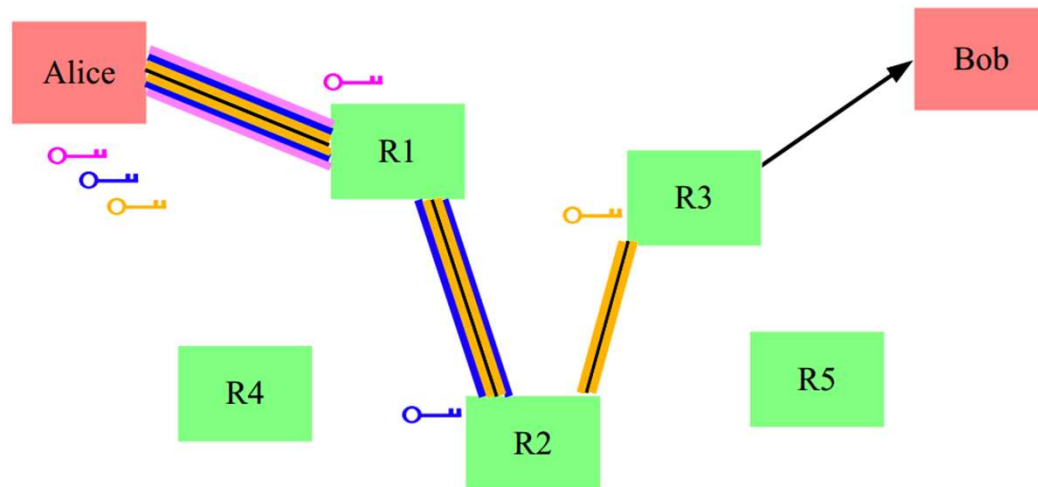
How Tor works



Safe(ish) if at least one router honest

Key challenge: hiding routing information

Solution: layered encryption



Side effect: contents encrypted from Alice to exit node

BUT: Unencrypted from exit node to Bob

Hidden services

What if the server wants to hide its address?

Simplified:

1. Connect to “rendezvous point” through Tor
2. Publish name → rendezvous point mapping
3. Client connects to rendezvous point

Onion address looks like **`http://3g2upl4pq6kufc4m.onion/`**

Silk Road

- Communication: Tor hidden service
- Payment: Bitcoin
- Security?
- Anonymous shipping?